

Values, Dynamic Typing and Desugaring

CS 350

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Introducing Boolean

Desugaring

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- We want our interpreter to have multiple *types*
 - Numbers and Booleans

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- Recall: implementation language features are *separate* from features of language being implemented

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 - E.g. Python, Ruby, JavaScript, PHP, Racket, Lua
- Recall: implementation language features are *separate* from features of language being implemented
- We can implement a dynamically-typed language (Curly-Bool) in a statically typed language (Flit)


```
<expr> ::=  
| <number>  
| <boolean>  
| {+ <expr> <expr>}  
| {* <expr> <expr>}  
| {if <expr> <expr> <expr>}  
| {zero? <expr> }
```

- Allow boolean literals

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- if instead of if0

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| {* <expr> <expr>}  
| {if <expr> <expr> <expr>}  
| {zero? <expr> }
```

- Allow boolean literals
- if instead of if0
- One operation producing a boolean

Example Programs

Example Programs

```
{if {zero? {+ {* 40 50}}  
    {* -1 2000}}  
  1  
 -1}
```

Representing Booleans

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```
(define-type Value  
  (numV [n : Number])  
  (boolV [b : Boolean]))
```

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- The constructor records run-time type information

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Abstract Syntax

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 - Add E to names for Expression
 - V was for Value

```
(define-type Exp
  (numE [n : Number])
  (boolE [b : Boolean])
  (plusE [left : Exp
           [right : Exp]])
  (timesE [left : Exp
            [right : Exp]])
  (cndE [test : Exp
         [thenCase : Exp]
         [elseCase : Exp]]))
```

Following the Types

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```
(define (interp [e : Exp] ) : Value
  (type-case Exp e
    ...
    [(plusE l r)
     (+ (interp l) (interp r))] ;; TYPE ERROR
```

- (interp l) and (interp r) produce Value

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- + expects Number

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```
(define (lift-binop [op : (Number Number -> Number)]
               [v1 : Value]
               [v2 : Value])
  : Value
  (type-case Value v1
    [(numV n1)
     (type-case Value v2
       [(numV n2)
        (numV (op n1 n2))]
       [else (error 'lift-binop "expects RHS to be a number")])]
    [else (error 'lift-binop "expects LHS to be a number")]))
```

Using the Lifting

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```
(define (interp [e : Exp] ) : Value
  (type-case Exp e
    ...
    [(plusE l r)
     (lift-binop + (interp l) (interp r))]
    [(timesE l r)
     (lift-binop * (interp l) (interp r))])
```

- Using a higher-order function means we can use for `plus` and `times`

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- Using a higher-order function means we can use for `plus` and `times`
- Textbook just defines `add` operation that works on values

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- Otherwise, same as `if0`

```
;; Evaluate Expressions
(define (interp [e : Exp] ) : Value
  (type-case Exp e
    ...
    [(cndE test thn els)
     (type-case Value (interp test)
       [(boolV b)
        (if b
            (interp thn)
            (interp els))]
       [else (error 'interp "Non-boolean given to if")])])])
```

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(define (interp [e : Exp] ) : Value
  (type-case Exp e
    ...
    [(zero?E e)
     (type-case Value (interp e)
       [(numV n)
        (boolV (= n 0))]
       [else
        (error 'interp "Expected number")]]])
```

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Design Decision

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 - `#t` and `#f` are not zero, so we could produce false
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 - e.g. in Untyped Racket, `#f` is false and everything else is true

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 - e.g. try it out

Desugaring

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- Some language features aren't strictly necessary, but are nice to have
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- These “nice to have” are called *syntactic sugar*
 - They “sweeten” the experience of programming

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for (int i = 0; i < 10; i++){  
    f();  
}  
// same as  
int i = 0;  
while (i < 10){  
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 - Simplifies the core language

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 - The result of desugaring
 - The type of tree we give to `interp` to actually run

Subtraction: Datatype

- First need the surface AST

Subtraction: Datatype

- First need the surface AST

Subtraction: Datatype

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```
(define-type ExpS
  (numS [n : Number])
  (plusS [left : ExpS]
         [right : ExpS])
  (timesS [left : ExpS]
          [right : ExpS])
  (cndS [test : ExpS]
        [thenBranch : ExpS]
        [elseBranch : ExpS])
  (subS [left : ExpS]
        [right : ExpS]))
```

- Exact same as core AST, but with subtraction

Subtraction: Parser Case

- Parser now produces a surface expression

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```
(define (parse [s : S-Exp]) : ExpS
  ...
  [(s-exp-match? `{- ANY ANY} s)
   (SurfSub (parse (second (s-exp->list s)))
             (parse (third (s-exp->list s))))])]
```


Desugaring: Core Cases

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Desugaring: Core Cases

- Define function that translates from `ExpS` to `Exp`

```
(define (desugar [es : ExpS]) : Exp
  (type-case ExpS es
    [(numS n)
     (numE n)]
    [(boolS b)
     (boolE b)]
    [(plusS l r)
     (plusE (desugar l) (desugar r))]
    [(timesS l r)
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    ...
```

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 - Pattern match
 - Recursively desugar the parts
 - Re-assemble with the corresponding Exp constructor

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- Example:

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```
(define (subE [l : Exp]
              [r : Exp])
  : Exp
  (plusE l
        (timesE (numE -1) r)))
```

- e.g. $\{- x y\}$ becomes $\{+ x \{* -1 y\}\}$

Desugaring: Subtraction Helper Function

- There is no `subE` in the `Exp` definition
 - Could add it, but don't need to
 - Instead, translate it into *existing* features of `Exp`
 - e.g. Our *core syntax*

```
(define (subE [l : Exp]
              [r : Exp])
  : Exp
  (plusE l
        (timesE (numE -1) r)))
```

- e.g. $\{-\ x\ y\}$ becomes $\{+\ x\ \{\ast\ -1\ y\}\}$
- Note: `subE` is a function, *not* a constructor

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(define (desugar [es : ExpS]) : Exp
  (type-case ExpS es
    ...
    [(subS l r)
     (subE (desugar l) (desugar r))])
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- Why did we desugar `l` and `r`?

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 - Can't give to `plusE` until they're expressions
 - e.g. what if `-` occurs deep in an expression

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- Simplifies the interpreter
 - `interp` is usually the most complicated part

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- Compilation is just a bunch of desugaring passes
 - Simpler and simpler languages until we have something simple enough for assembly code

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```
(define-type ExpS
  ...
  (eqS [l : ExpS] [r : ExpS]))

(define (eqE [l : Exp]
             [r : Exp])
  : Exp
  (zero?E (subE l r)))

(define (desugar [es : ExpS]) : Exp
  ...
  [(eqS l r)
   (eqE (desugar l) (desugar r))])
```

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```
(define (andE [l : Exp]
              [r : Exp])
  : Exp
  (cndE l
        r
        (boolE #f)))
```

```
(define (orE [l : Exp]
             [r : Exp])
  (cndE l (boolE #t) r))
```

```
(define (notE [e : Exp])
  (cndE e (boolE #f) boolE #t))
```